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Generating Land Use Land Cover (LULC) Map using Supervised Classification Techniques

Prepared By: Satwik Shreshth
Programme: MCA 2nd Year
University: Sikkim University (Central University)
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Abstract

Land use and land cover (LULC) mapping is a fundamental task in earth observation that characterises the physical state of the land surface and the human activities superimposed upon it. This report documents a cloud-computing-based LULC mapping study for East Sikkim district, Sikkim, India, conducted as part of the Summer Training Programme on Remote Sensing and GIS – 2025 at the India Space Academy. East Sikkim is an ecologically complex Himalayan district spanning approximately 1,521.8 km², characterised by extreme altitudinal variation, dense natural forests, active agricultural terraces, rapidly expanding urban settlements, and high-altitude snow and glacial zones.

Multi-temporal Sentinel-2 Surface Reflectance imagery (January 2020 – January 2024) was accessed and processed on the Google Earth Engine (GEE) cloud computing platform. A custom cloud-masking routine incorporating snow-pixel preservation was applied, and a multi-year median composite was generated. A 15-feature input stack was constructed by combining six Sentinel-2 spectral bands, three spectral indices (NDVI, NDWI, NDSI), three terrain derivatives from the USGS SRTM digital elevation model, and three phenological NDVI features that capture the seasonal contrast between monsoon and winter vegetation states.

Two supervised ensemble classifiers – Random Forest (RF) and Gradient Tree Boosting (GTB) – were trained on 883 labelled samples and evaluated on a held-out validation set of 425 samples across five thematic classes: Water, Forest, Highland / Snow Cover, Built-up Area, and Barren Land. The RF classifier was used to generate the final 10-m resolution LULC map, exported as a GeoTIFF and visualised in QGIS.

Keywords: LULC, Google Earth Engine, Sentinel-2, Random Forest, Gradient Tree Boosting, East Sikkim, Himalaya, supervised classification, remote sensing.

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I. Introduction

Land use and land cover (LULC) characterises the physical attributes of the Earth’s surface and the anthropogenic activities imposed upon it. Spatially explicit LULC information is essential for forest carbon accounting [1], urban heat-island analysis [2], agricultural monitoring [3], hydrological modelling [4], and biodiversity assessment [5].

The Eastern Himalaya, and Sikkim in particular, exemplifies the challenge of LULC mapping in mountain terrain. The state encompasses exceptional biological diversity, with a vegetation gradient from subtropical forests in the south to alpine meadows and glaciated peaks in the north [6]. East Sikkim – the most densely populated district and the seat of the state capital, Gangtok – is experiencing rapid urbanisation, horticulture-driven land-use transitions, and significant cryosphere changes [7]. Accurate, current LULC data are therefore indispensable for informed land management.

Cloud-computing platforms, most notably Google Earth Engine (GEE), have transformed the feasibility of satellite-based LULC classification at regional scales [8]. The Sentinel-2 constellation, with its 10-m spatial resolution and five-day revisit cycle [9], combined with ensemble machine-learning classifiers such as Random Forest [10] and Gradient Tree Boosting [11], provides both the data quality and analytical power required for accurate land-cover monitoring in complex terrain.

This report presents an end-to-end workflow for generating a 10-m LULC map of East Sikkim using GEE, comparing RF and GTB classifiers, and visualising the final map in QGIS.

II. Study Area

East Sikkim is one of the four administrative districts of Sikkim state in north-eastern India (Figure 2). The district spans latitudes 27.14°N – 27.42°N and longitudes 88.44°E – 88.92°E, covering approximately 1,521.8 km² as delineated by the FAO GAUL 2015 Level-2 boundary. The district headquarters, Gangtok, lies at approximately 1,650 m above mean sea level.

The terrain spans altitudes from ~300 m in the southern foothills to over 4,000 m on northern ridgelines, creating a corresponding spectrum of vegetation communities: sub-tropical broadleaved forest → temperate mixed forest → sub-alpine scrub → alpine meadow. The Teesta River and its tributaries have carved deeply incised valleys with steep slopes that complicate spectral discrimination from satellite data.

The climate is governed by a strong South Asian monsoon (June–September), delivering over 70% of annual precipitation. Mean annual rainfall in Gangtok exceeds 3,500 mm. Winters (December–February) are dry with snowfall at higher elevations. Land use is a mosaic of dense natural forest, cardamom and tea cultivation, rice and vegetable

cultivation on terraces, scattered settlements, and exposed rock or snowfields at higher altitudes.

III. Data and Methodology

A. Satellite Data and Pre-processing

All processing was performed within GEE (project: lulc-467806). The Sentinel-2 Surface Reflectance Harmonised collection (COPERNICUS/S2_SR_HARMONIZED) was accessed for 1 January 2020 to 31 January 2024 with a first-pass cloud-cover threshold of 80%. A per-image cloud-masking function was then applied using the QA60 bitmask (bits 10 and 11 for opaque cloud and cirrus). A snow-preservation step retained pixels with Normalised Difference Snow Index (NDSI) > 0.40 regardless of cloud flags:

$$\text{mask} = \text{CloudFree} \cup (\text{NDSI} > 0.40) \quad (1)$$

After scaling to surface reflectance (divide by 10,000), a pixel-wise median composite was generated across the full collection and clipped to the AOI.

B. Feature Engineering

The classification input comprised 15 predictor features (Table 1): six Sentinel-2 spectral bands, six spectral and phenological indices, and three terrain derivatives from the USGS SRTM 30-m DEM. The phenological features – NDVI computed separately for the winter period (December 2021 – February 2022) and monsoon period (June – September 2021), and their arithmetic difference – capture the seasonally distinct spectral behaviour of vegetation types. Deciduous crops and highland grasslands exhibit a large monsoon-to-winter NDVI drop, whereas evergreen forest responds modestly, providing an orthogonal discriminant axis.

Table 1: Feature stack used for LULC classification (15 features).

Category	Feature	Formula / Source	Purpose
Spectral bands	B2 (Blue, 490 nm)	Sentinel-2 SR	Surface reflectance
	B3 (Green, 560 nm)	Sentinel-2 SR	Surface reflectance
	B4 (Red, 665 nm)	Sentinel-2 SR	Surface reflectance
	B8 (NIR, 842 nm)	Sentinel-2 SR	Vegetation / water
	B11 (SWIR-1, 1610 nm)	Sentinel-2 SR	Moisture / soil
	B12 (SWIR-2, 2190 nm)	Sentinel-2 SR	Geology / soil
Indices	NDVI	$(B8 - B4)/(B8 + B4)$	Vegetation density
	NDWI	$(B3 - B8)/(B3 + B8)$	Open water
	NDSI	$(B3 - B11)/(B3 + B11)$	Snow / ice
	NDVI _{win}	$(B8 - B4)/(B8 + B4)$, Dec-Feb	Winter vegetation
	NDVI _{mon}	$(B8 - B4)/(B8 + B4)$, Jun-Sep	Monsoon vegetation
	NDVI _{diff}	NDVI _{mon} - NDVI _{win}	Phenological variability
Terrain	Elevation	USGS SRTMGL1 (30 m)	Altitudinal zone
	Slope	Derived from DEM	Topographic steepness
	Aspect	Derived from DEM	Solar exposure

C. Land Cover Classification Scheme

Five thematic classes were defined to balance ecological relevance and spectral separability (Table 2). Class labels are consistent with the QGIS map legend.

Table 2: LULC classification scheme and ESA WorldCover correspondence.

ID	Class Name	WorldCover Codes	Description
0	Water	80, 90	Rivers, lakes, reservoirs, permanent / seasonal water bodies
1	Forest	10, 20	Dense forest, closed-canopy tree cover
2	Highland / Snow Cover	30, 40, 80, 90	Alpine shrubland, herbaceous vegetation, seasonal snow, glaciers
3	Built-up Area	50	Urban areas, settlements, impervious surfaces
4	Barren Land	60	Bare rock, soil, scree, sparse or absent vegetation

D. Training Data

Training data were assembled from two sources:

1. Manual ground-truth points (346) digitised against high-resolution Google Satellite imagery within GEE (asset: projects/lulc-467806/assets/training_gangtok).

2. ESA WorldCover 2021 stratified samples (1,000) – 200 points per class extracted automatically from the remapped WorldCover product.

After merging and removing null observations, 1,337 labelled points were obtained. Feature values were sampled at 10-m scale (sampleRegions, tileScale=2), yielding 1,308 fully attributed samples. A 70/30 random split (seed = 42) produced 883 training and 425 validation samples. Per-class counts are given in Table 3 and illustrated in Figure 1.

Table 3: Training sample distribution across five LULC classes.

Class	Count	%
Water	250	18.7
Forest	277	20.7
Highland / Snow Cover	250	18.7
Built-up Area	299	22.4
Barren Land	261	19.5
Total	1,337	100.0

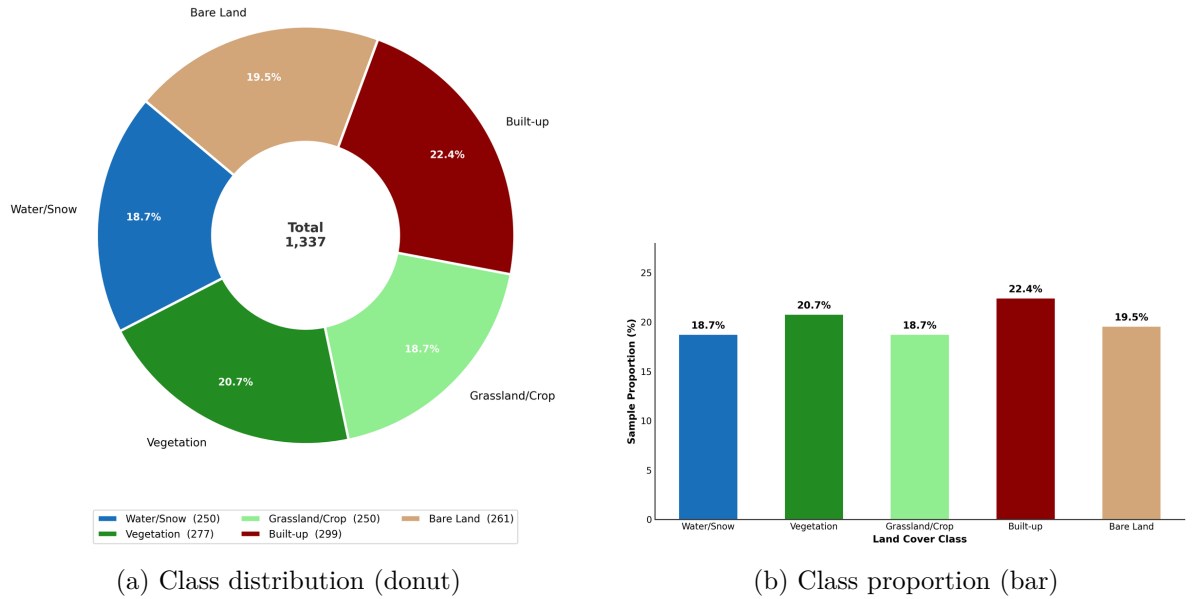


Figure 1: Training sample distribution. Total: 1,337 (346 manual + 1,000 WorldCover-derived).

E. Classifier Training

Two ensemble classifiers were trained using GEE’s SMILE library.

1) Random Forest (RF)

RF [10] builds an ensemble of decision trees on bootstrapped subsamples. Hyperparameters: 200 trees, 3 variables per split ($\approx \sqrt{15}$), minimum leaf population = 2, seed = 42.

2) Gradient Tree Boosting (GTB)

GTB [11] trains trees sequentially to correct the residual errors of the previous ensemble. Hyperparameters: 200 trees, shrinkage = 0.05, seed = 42.

F. Accuracy Assessment

Both classifiers were evaluated on the 425-sample validation set. Metrics computed from the GEE errorMatrix:

- Overall Accuracy (OA): fraction of correctly classified samples.
- Cohen’s Kappa (κ): chance-corrected agreement.
- Producer’s Accuracy (PA): recall per class.
- Consumer’s Accuracy (CA): precision per class.
- F1-score: $F_1 = 2 \text{ PA} \cdot \text{CA} / (\text{PA} + \text{CA})$.
- AUC-ROC & Average Precision (AP): from MULTIPROBABILITY outputs using scikit-learn [12].

G. Map Production

RF was selected for final map export. The classified 10-m raster was exported as a GeoTIFF (WGS84, EPSG:4326) via `Export.image.toDrive` and visualised in QGIS with a five-class colour scheme.

IV. Results

A. LULC Map of East Sikkim

The RF-derived LULC map is shown in Figure 2. Forest dominates the middle-elevation forested slopes; Highland / Snow Cover occupies valley floors, terraces, and sub-alpine zones; Built-up Area is clustered around Gangtok and road corridors; Water and Barren Land are confined to the north-eastern high-altitude zones and riverine strips.

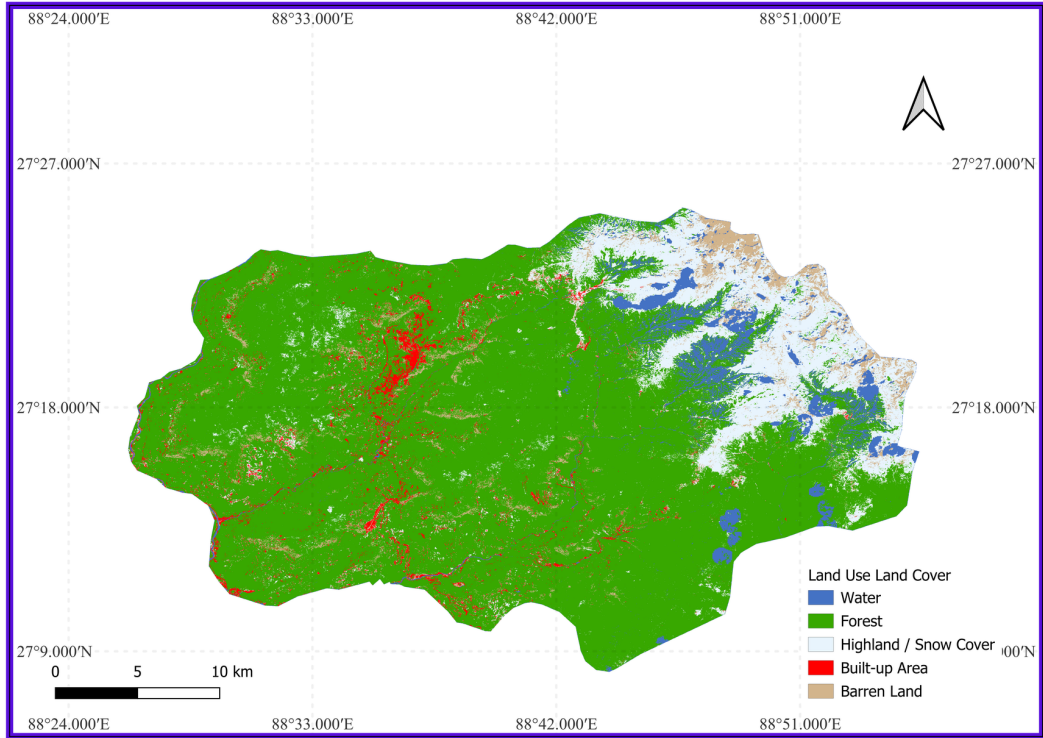


Figure 2: LULC map of East Sikkim at 10-m resolution (Random Forest, QGIS visualisation). Classes: Water (blue), Forest (dark green), Highland / Snow Cover (light green), Built-up Area (red), Barren Land (tan). Coordinate system: WGS84 / EPSG:4326.

B. Overall Classification Accuracy

Table 4 and Figure 3 present the headline accuracy metrics. GTB marginally outperforms RF by 0.24 percentage points in OA and 0.0034 in κ . Both Kappa values exceed 0.60, confirming substantial agreement beyond chance [13]. Macro-averaged AUC-ROC values of 0.9414 (RF) and 0.9481 (GTB) confirm excellent class separability in probability space.

Table 4: Headline accuracy metrics for RF and GTB.

Classifier	OA (%)	Kappa (κ)	Macro AUC-ROC
Random Forest (RF)	75.76	0.6956	0.9414
Gradient Tree Boosting (GTB)	76.00	0.6990	0.9481

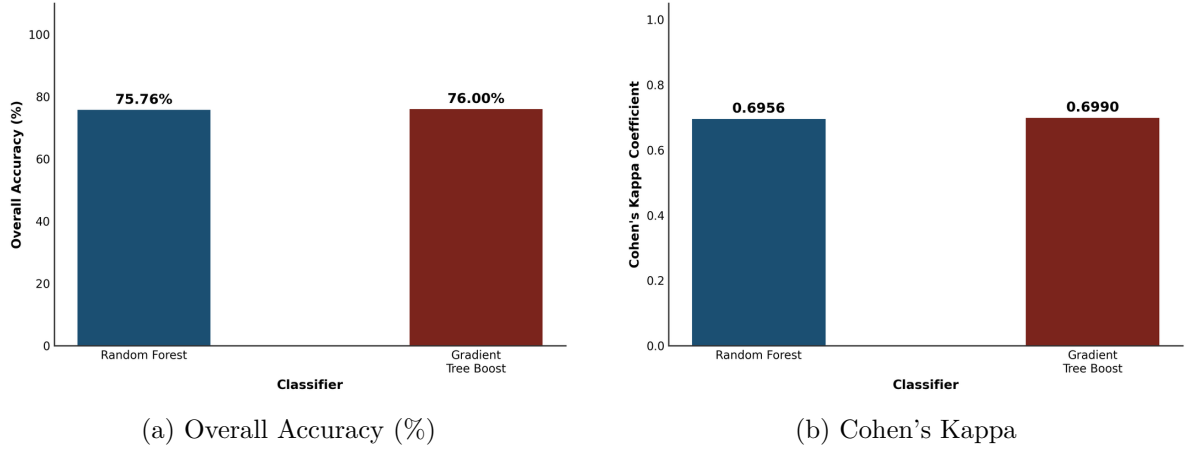


Figure 3: Overall Accuracy and Cohen's Kappa: RF vs GTB.

C. Confusion Matrices

Tables 5 and 6 present the 5×5 confusion matrices; Figure 4 shows the heatmap visualisations. The dominant source of error is the confusion of Highland / Snow Cover with Forest (21 samples for both classifiers), reflecting spectral overlap between herbaceous vegetation and open forest understorey, particularly at 10-m pixel scale along terrace–forest boundaries.

Table 5: Confusion matrix – Random Forest ($n = 425$). Rows: actual; columns: predicted.

	Snow	Veg	High	Blt	Bare
Water	74	12	2	4	3
Forest	1	77	1	0	0
Highland / Snow Cover	2	21	45	6	7
Built-up Area	0	6	4	89	0
Barren Land	9	14	3	8	37

Table 6: Confusion matrix – Gradient Tree Boosting ($n = 425$). Rows: actual; columns: predicted.

	Snow	Veg	High	Blt	Bare
Water	72	10	2	3	8
Forest	2	72	4	0	1
Highland / Snow Cover	1	21	47	6	6
Built-up Area	0	5	4	88	2
Barren Land	9	9	4	5	44

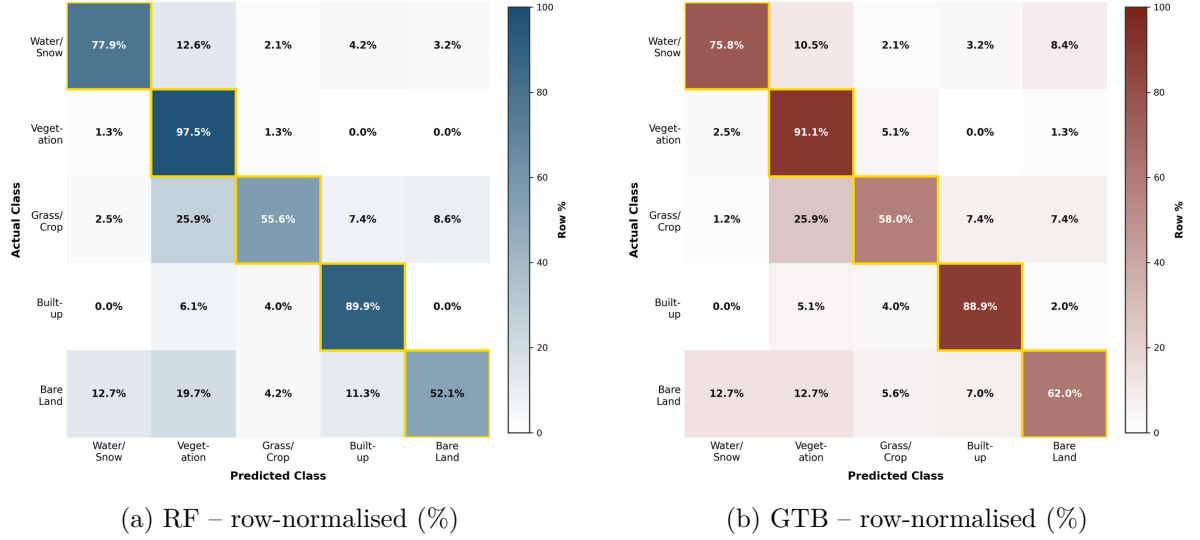


Figure 4: Row-normalised confusion matrices showing the percentage of each actual class assigned to each predicted class. High diagonal values indicate accurate per-class discrimination; off-diagonal entries reveal misclassification patterns.

D. Per-class Accuracy Metrics

Table 7 presents the full per-class breakdown; Figure 5 compares F1-scores, and Figure 6 shows omission and commission errors.

Table 7: Per-class accuracy metrics. PA = Producer’s Accuracy; CA = Consumer’s Accuracy; F1 = F1-score; OE = Omission Error; CE = Commission Error. All values in %.

Class	PA		CA		F1		OE		CE	
	RF	GTB	RF	GTB	RF	GTB	RF	GTB	RF	GTB
Water	77.89	75.79	86.05	85.71	81.77	80.45	22.11	24.21	13.95	14.29
Forest	97.47	91.14	59.23	61.54	73.68	73.47	2.53	8.86	40.77	38.46
Highland / Snow Cover	55.56	58.02	81.82	77.05	66.18	66.19	44.44	41.98	18.18	22.95
Built-up Area	89.90	88.89	83.18	86.27	86.41	87.56	10.10	11.11	16.82	13.73
Barren Land	52.11	61.97	78.72	72.13	62.71	66.67	47.89	38.03	21.28	27.87

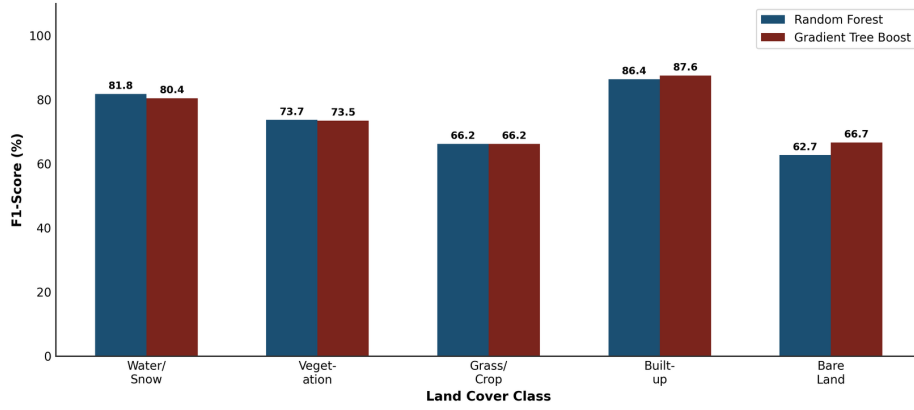


Figure 5: F1-score comparison per class – RF vs GTB. Built-up Area achieves the highest F1 (>86%) for both classifiers. GTB shows an advantage of 3.96 points for Barren Land; RF achieves higher recall for Forest.

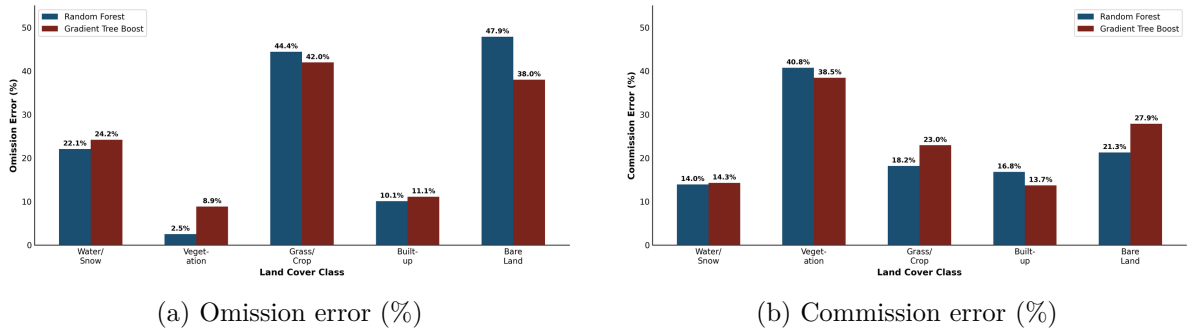


Figure 6: Omission and commission errors per class for RF and GTB. Highland / Snow Cover and Barren Land exhibit the highest omission errors (>40%), indicating frequent misclassification into adjacent spectral classes.

E. AUC-ROC and Precision–Recall Analysis

Table 8 presents per-class AUC-ROC and Average Precision scores. Figure 7 shows the per-class ROC curves, and Figure 8 the macro-average comparison.

Table 8: Per-class AUC-ROC and Average Precision (AP) for RF and GTB.

Class	AUC-ROC		Avg. Precision	
	RF	GTB	RF	GTB
Water	0.9526	0.9556	0.9008	0.9009
Forest	0.9370	0.9498	0.7217	0.7786
Highland / Snow Cover	0.9207	0.9194	0.7812	0.7709
Built-up Area	0.9756	0.9767	0.9458	0.9528
Barren Land	0.9209	0.9391	0.7416	0.7777
Macro	0.9414	0.9481	—	—

All five classes exceed $AUC > 0.92$ for both classifiers, confirming strong probabilistic discriminability well above the hard-label OA. Built-up Area achieves the highest AUC

(0.9756 RF, 0.9767 GTB) and AP (>0.94), consistent with its distinctive SWIR and negative-NDVI signature. The lower AP for Forest (0.72/0.78) and Highland / Snow Cover (0.78/0.77) relative to their AUC values reflects class imbalance and inter-class confusion along vegetation–grassland boundaries.

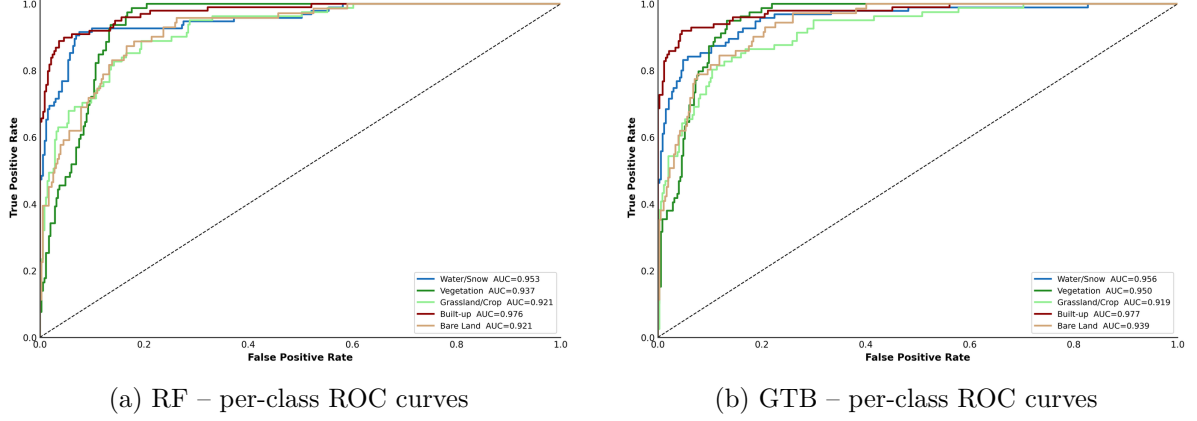


Figure 7: ROC curves for each LULC class. AUC values are shown in the legend. All classes exceed $AUC > 0.92$ for both classifiers.

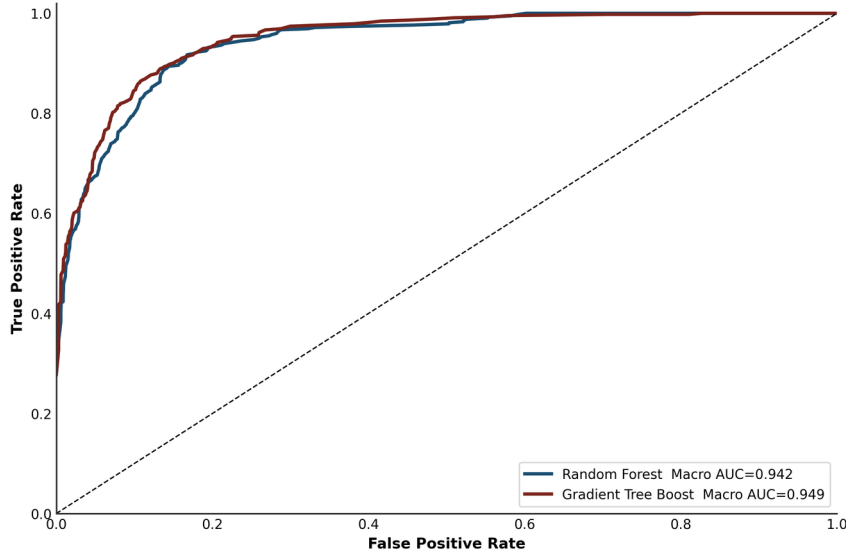


Figure 8: Macro-averaged ROC curve: RF (AUC = 0.9414) vs GTB (AUC = 0.9481).

V. Discussion

A. Classifier Comparison

The near-identical OA values (RF: 75.76%, GTB: 76.00%) indicate that the 15-feature input stack, rather than classifier choice, is the primary driver of performance. However, GTB’s sequential boosting mechanism confers a meaningful per-class advantage for Barren Land, improving recall by 9.86 percentage points and F1 by 3.96 points relative to RF. The consistently higher AUC-ROC of GTB across four of five classes

further supports its marginal superiority in probabilistic discrimination, consistent with comparable Himalayan LULC studies [14].

B. Sources of Misclassification

The dominant confusion is between Highland / Snow Cover and Forest (21 misclassified samples in both matrices). Managed agricultural plots on terraced slopes share overlapping spectral signatures with open forest understorey, particularly at 10-m pixel scale where spectral mixing is substantial. The $\text{NDVI}_{\text{diff}}$ feature partially mitigates this through phenological contrast, but it cannot fully resolve sub-pixel mixing along terrace–forest boundaries.

High-altitude confusion between Barren Land and Highland/Snow Cover reflects the spectral similarity of dry, bright rock surfaces and patchy snowpack in NDSI and SWIR bands. Elevation and aspect features partially stratify this confusion; incorporation of Sentinel-1 SAR backscatter in future work would further help. The large commission error for Forest (40.77% RF, 38.46% GTB) indicates substantial false-positive assignments from Highland / Snow Cover pixels during the monsoon season.

C. Role of Phenological and Terrain Features

The three phenological features – NDVI_{mon} , NDVI_{win} , and $\text{NDVI}_{\text{diff}}$ – contributed an orthogonal discriminant axis to the median spectral bands. The seasonal NDVI contrast is large for Highland / Snow Cover (cropland/alpine meadow) and small for evergreen Forest, aiding their separation. Terrain features (elevation, slope, aspect) provided ecologically grounded stratification, particularly for distinguishing Highland/Snow Cover and Barren Land at high altitudes from spectrally similar low-altitude classes. Phenological enrichment of feature stacks has been demonstrated to improve classification accuracy in agricultural and montane landscapes [15, 16].

D. Limitations

1. An OA of $\sim 76\%$ is moderate for a five-class scheme over complex Himalayan terrain at 10-m resolution and reflects inherent spectral mixing.
2. WorldCover-derived training samples may propagate label noise from that product, particularly in transition zones.
3. No post-classification spatial smoothing was applied; salt-and-pepper noise is evident in heterogeneous zones.
4. Persistent monsoon cloud cover introduces potential temporal bias in the median composite at high-altitude pixels.
5. Area estimates from pixel counts have not been bias-corrected for map accuracy [17].

VI. Conclusion

This study presented a reproducible, cloud-computing-based workflow for generating a 10-m LULC map of East Sikkim, India, integrating multi-temporal Sentinel-2 imagery, USGS SRTM terrain derivatives, and seasonal NDVI phenological features within Google Earth Engine. Five thematic classes – Water, Forest, Highland/Snow Cover, Built-up Area, and Barren Land – Highland / Snow Cover, Built-up Area, and Barren Land – were mapped using a hybrid training dataset of 1,337 samples assembled from manual field annotations and stratified ESA WorldCover 2021 samples.

Random Forest and Gradient Tree Boosting classifiers achieved comparable overall accuracies ($\sim 76\%$) and strong macro-averaged AUC-ROC (>0.94), with GTB offering marginal advantages in probabilistic discrimination and Barren Land recall. Built-up Area was the most accurately mapped class ($F1 > 86\%$), while Highland / Snow Cover and Barren Land exhibited the highest misclassification rates due to spectral overlap with Forest and Highland/Snow Cover, respectively. The final LULC map provides a baseline characterisation of East Sikkim’s land cover suitable for district-level land management, carbon stock estimation, and urban growth monitoring.

Future work should explore SAR-optical data fusion for improved snow/bare discrimination, object-based analysis to reduce noise, deep-learning approaches exploiting spatial context, and bias-corrected area estimation following Olofsson et al. [17].

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